
Pattern Arithmetic: Phase-Colored Geometric Framework for High-Density Information via Unit Probability Spheres

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Abstract

Traditional matrix representations of continuous state spaces suffer from severe structural limitations: rigid coordinate axis dependencies, exponential spatial scaling costs ($O(N^3)$), and the destructive overwrite of overlapping signals at identical spatial coordinates. To overcome these constraints, this paper introduces *Pattern Arithmetic*, a continuous geometric framework that models multi-variable information states using state vectors on a Unit Probability Sphere $\mathcal{S} \subset \mathbb{R}^3$, augmented by an internal phase-color degree of freedom $\chi \in [0, 2\pi)$. By mapping continuous probability distributions over vector fields and treating color as an independent phase factor, overlapping states occupy identical spatial coordinates without loss of information density. We establish six foundational operational axioms, including Information Idempotence ($1 \oplus 1 = 1$), Incremental Subsumption ($A + A_{0.01} = A_{0.01}$), and Phase Interference ($A - A = 0$). Finally, we demonstrate how this pattern-based algebra provides a parameter-efficient alternative for high-density latent storage, signal unmixing, and physical field modeling.

1 Introduction

Ordinary arithmetic uses the integer 1 as a unit of truth. The identity $1 = 1$ says that two tokens are of the same kind. The rule $1 + 1 = 2$ says that combining two such tokens produces a larger quantitative state. That algebra answers the question *how many?*

It is not the only algebra. A color and a musical note are also units of truth, but their combination does not enlarge a pile. Under additive light, red and green yield yellow. Two copies of the same yellow, once brightness is normalized, remain yellow. Two notes

yield a chord, not a larger integer; a tone plus its opposite phase yields silence. In each case the word “add” names a different rule, and therefore a different kind of unit.

When the object of interest is a structured whole rather than a count, a hue, or a pitch, the natural unit is a *pattern*. This paper represents one pattern by one state on a unit probability sphere $\mathcal{S} \subset \mathbb{R}^3$:

- $r \in [0, 1]$ is the strength of the unit, the descendant of the numeral 1, capped so that a state remains a single unit of truth;
- (θ, ϕ) is the direction of the pattern on the sphere;
- $\chi \in [0, 2\pi)$ is an internal phase-color, so two identities may occupy the same direction without overwriting one another.

Pattern Arithmetic is the algebra of these units. Identical states do not double after normalization ($A \oplus A = A$). Opposite phases cancel ($A - A = 0$). Rotations of the sphere preserve the unit.

A single numerical picture is enough to see the difference from counting. Take two unit states at the same place, one with phase 0° (red) and one with phase 120° (green). Counting would return 2. Phase-aware combination returns a single unit whose blended phase is 60° (yellow). The same two strengths with opposite phase (0° and 180°) return the null state, black. Two units of truth become either one new identity or none.

1.1 The Limitations of Discretized Matrix Grids

A discrete matrix grid is the wrong box for such a unit. Traditional matrix representations of continuous state spaces suffer from two severe structural bottlenecks:

- **Memory Scaling Bottlenecks:** A voxel cube of resolution N stores $O(N^3)$ cells, growing steeply with resolution.
- **Signal Identity Destruction:** Mapping multiple overlapping variables to identical spatial coordinates often results in the destructive overwriting of underlying signal identities.

1.2 The Phase-Colored Unit Probability Sphere Framework

To resolve these grid constraints, the sphere is offered as one bounded object that can carry overlapping identities by direction and phase, without enlarging a lattice.

- **Spatial Orientation** (θ, ϕ) : Defines continuous directional coordinates across the spherical surface.
- **Magnitude Likelihood** $(0 \leq r \leq 1)$: Represents bounded probability density within the sphere.
- **Internal Color Phase** $(\chi \in [0, 2\pi))$: Serves as an independent phase degree of freedom.

1.3 Core Contributions & High-Level Intuition

The principal contributions of Pattern Arithmetic are:

- **High-Density Superposition:** Storing overlapping state vectors concurrently within a single bounded spatial manifold.
- **Parameter-Efficient Field Storage:** Eliminating discrete voxel bounds by treating states as continuous probability distributions.

The next section turns these definitions into formal mathematical axioms.

2 Axiomatic Foundations Of Pattern Arithmetic

2.1 Axiom 1: The Normalized Unit Truth State ($\mathcal{S} = 1$)

To ensure stability across multi-variable operations, every state vector field $\vec{v} = (r, \theta, \phi, \chi)$ on the Unit Probability Sphere $\mathcal{S} \subset \mathbb{R}^3$ represents an immutable unit of truth ($\mathcal{S} = 1$). Global likelihood integrates strictly to unity:

$$\iiint_{\mathcal{S}} P(r, \theta, \phi) dV = 1.0 \quad (1)$$

2.2 Axiom 2: The Absolute Null State ($\mathcal{S} = 0$, Black)

The state $\mathcal{S} = 0$ defines the absolute lower bound of the state space, corresponding to complete destructive interference, total phase cancellation, or zero-state equilibrium:

$$\|\vec{v}\| = 0 \quad \forall (\theta, \phi, \chi) \implies \mathcal{S} = 0 \quad (\text{Black}) \quad (2)$$

Physically and visually, $\mathcal{S} = 0$ represents an empty probability field devoid of active signal energy or directional orientation.

2.3 Axiom 3: The Infinite Superposition State ($\mathcal{S} = \infty$, White)

The state $\mathcal{S} = \infty$ defines the theoretical upper bound of maximal entropy, representing total spectral superposition across all continuous orientations (θ, ϕ) and internal color-phase frequencies $\chi \in [0, 2\pi)$:

$$\mathcal{S} = \lim_{N \rightarrow \infty} \bigoplus_{i=1}^N A_i \implies \mathcal{S} = \infty \quad (\text{White}) \quad (3)$$

Physically and visually, $\mathcal{S} = \infty$ corresponds to full-spectrum white light, where all possible state vectors co-exist simultaneously without spatial exclusion.

2.4 Axiom 4: Information Idempotence ($1 \oplus 1 = 1$)

Combining identical information states yields no expansion of likelihood volume. Upon intensity normalization:

$$A \oplus A = \frac{A + A}{\|A + A\|} = A \quad (4)$$

2.5 Axiom 5: Phase-Aware Superposition & Interference

The combination of two truth states A and B is governed by relative phase difference $\Delta\chi$:

$$A \pm B = \sqrt{\|A\|^2 + \|B\|^2 + 2\|A\|\|B\|\cos(\Delta\chi)} \quad (5)$$

Destructive anti-phase interference ($\Delta\chi = 180^\circ$) collapses composite states directly into the Null State ($A - A = 0$, Black) defined in Axiom 2.2.

2.6 Axiom 6: Rotational Conservation ($\times \mathbf{R}$)

Multiplication operates as a Lie group rotation $\mathbf{R} \in \text{SO}(3)$ in continuous spherical space, preserving total truth volume:

$$\|\mathbf{R} \cdot A\| = \|A\| = 1.0 \quad (6)$$

3 Transformation Algebra: From Identity to Expression

3.1 Motivation: Two Layers of Meaning

The operators established so far ($\oplus, \pm$) answer the question *what is this state?* Combining Sky, Sun, and Wind under $\oplus$ yields a single, order-independent resultant — a new composite unit of truth, exactly as combining Ruler and Man yields the composite unit King. This is a *lexical* operation: it mints identities, the way a dictionary mints words from morphemes. It does not, and structurally cannot, distinguish a poem in which the sun rises through a wind-blown sky from a poem in which wind extinguishes a darkening sun — both share the identical resultant $R = \text{Sky} \oplus \text{Sun} \oplus \text{Wind}$, since $\oplus$ is commutative and idempotent by Axioms 2.4 and 2.5.

The distinction between these two poems is not a difference in *ingredients*; it is a difference in *how one state acts upon another, and in what order*. This is a *relational* operation, and it requires a second operator, T , layered on top of $\oplus$: where $\oplus$ builds nouns, T builds sentences. A poem is not the resultant R itself, but a specific trajectory traced across $\mathcal{S}$ by successive transformations applied to R .

3.2 Formal Definition of the Transformation Operator

Rather than introduce T as an unrelated new primitive, it is defined using the same rotational machinery already given in Axiom 2.6: every state $w \in \mathcal{S}$ acts on another state $x \in \mathcal{S}$ by rotating x 's direction about w 's own axis, by an angle proportional to w 's strength, while additively shifting x 's phase by w 's phase:

$$T_w(x) = \left(r_x, \text{Rot}_{\hat{n}(\theta_w, \phi_w)}(\alpha(r_w)) \cdot (\theta_x, \phi_x), (\chi_x + \chi_w) \bmod 2\pi \right) \quad (7)$$

where $\hat{n}(\theta_w, \phi_w)$ is the unit axis defined by w 's direction, $\text{Rot}_{\hat{n}}(\alpha) \in \text{SO}(3)$ is the rotation of Axiom 2.6 about that axis by angle α , and $\alpha(r_w) = \pi r_w$ is the simplest monotonic choice mapping strength to rotation angle (a full-strength unit, $r_w = 1$, rotates by the maximal 180°). Magnitude r_x is left invariant: a transformation reorients and recolors a state, it does not strengthen or weaken it — that remains the role of $\oplus$.

3.3 Derived Properties

The following properties follow from Equation (7) rather than being separately postulated.

Identity. If $w = \mathbf{0}$ (the Null State of Axiom 2.2), then $\alpha(0) = 0$ and the phase shift is 0, so $T_{\mathbf{0}} = \text{id}$.

Invertibility. Define $w^* = (r_w, \theta_w, \phi_w, -\chi_w)$. Then $T_{w^*}(T_w(x)) = x$ for all x : the same axis and angle undo the rotation, and the phase shift cancels. Every transformation has a well-defined inverse. Note that this is distinct from anti-phase interference ($A - A = 0$, Axiom 2.5), which annihilates a state; T_{w^*} instead *undoes a transformation* on a state that persists.

Associativity. Since each T_w is a function $\mathcal{S} \rightarrow \mathcal{S}$, composition of transformations is associative automatically, by the associativity of function composition; no additional axiom is required.

Non-commutativity. Rotations about distinct axes in $\text{SO}(3)$ generically do not commute: for a, b with distinct directions (θ, ϕ) ,

$$T_a(T_b(x)) \neq T_b(T_a(x)). \quad (8)$$

This is not an additional assumption; it is inherited directly from the well-established non-commutativity of $\text{SO}(3)$, already invoked by Axiom 2.6. It is this property, and only this property, that allows two poems built from the same resultant R to diverge: order of application is meaningful even though the underlying identity is not.

3.4 Poems as Transformation Trajectories

A poem π built from resultant R and an ordered sequence of transforming states $w_1, w_2, \dots, w_n$ is the trajectory

$$\pi = (R; w_1, \dots, w_n) : R \longrightarrow T_{w_1}(R) \longrightarrow T_{w_2}T_{w_1}(R) \longrightarrow \dots \longrightarrow T_{w_n} \dots T_{w_1}(R). \quad (9)$$

Two poems may share an identical resultant R (the same theme, the same “ingredients”) while tracing entirely different trajectories — this is the formal sense in which many poems arise from the same units of truth. Conversely, because rotation composition is many-to-one, two distinct trajectories may terminate at the same endpoint; a poem’s identity is therefore carried by its full path through $\mathcal{S}$, not by its final state alone.

3.5 Worked Numerical Example: Sun, Wind, and Sky

Let Sky be the state acted upon, with direction $(\theta, \phi) = (45^\circ, 45^\circ)$, i.e. Cartesian axis $(0.500, 0.500, 0.707)$. Let Sun define an axis along $(\theta, \phi) = (90^\circ, 0^\circ)$, i.e. $(1, 0, 0)$, with strength $r_{\text{Sun}} = 1.0$, giving rotation angle $\alpha_{\text{Sun}} = 180^\circ$. Let Wind define an axis along $(\theta, \phi) = (90^\circ, 90^\circ)$, i.e. $(0, 1, 0)$, with strength $r_{\text{Wind}} = 0.7$, giving $\alpha_{\text{Wind}} = 126^\circ$.

Order 1 — Sun acts first, then Wind (the sun rising through a wind-blown sky):

$$T_{\text{Wind}}(T_{\text{Sun}}(\text{Sky})) \approx (-0.866, -0.500, 0.011) \quad (10)$$

Order 2 — Wind acts first, then Sun (wind extinguishing a darkening sun):

$$T_{\text{Sun}}(T_{\text{Wind}}(\text{Sky})) \approx (0.278, -0.500, 0.820) \quad (11)$$

The two endpoints are clearly distinct, confirming Equation (8): identical primitives, identical resultant R , opposite structural readings. Notably, if $\chi_{\text{Sun}} = 0^\circ$ and $\chi_{\text{Wind}} = 90^\circ$, the accumulated phase shift is 270° in *both* orderings, since phase addition is commutative. The two poems therefore share the same blended color/mood even as their spatial — structural — meaning diverges. This asymmetry (shared affect, divergent structure) is a direct and testable prediction of the model.

3.6 Scope

T is deliberately left unconstrained beyond Equation (7): any sequence of any transforming states is a formally valid trajectory. Pattern Arithmetic in this form supplies the substrate on which expression is built — the combinatorial space of possible trajectories — and does not yet supply a grammar constraining which trajectories are coherent, well-formed, or meaningful to a human reader. Whether such constraints belong inside this framework (as further axioms on admissible sequences of T) or outside it (as a learned or externally specified grammar layered on top) is left open.

4 Operational Algebra & Numerical Demonstration

4.1 Vector Addition & Phase Blending

Using Equation (5) from Axiom 2.5, combining Red Vector (0°) and Blue Vector (120°) at coordinate $(0.6, 0.8)$ results in deterministic phase blending ($\chi = 66.6^\circ$).

4.2 Rotational Transposition (Multiplication) & Unmixing (Division)

- **Multiplication** ($\times$): State transformation via Lie group rotation matrix $\mathbf{R} \in \text{SO}(3)$.
- **Division** ($\div$): Deconvolution and signal unmixing via inverse transformation $\mathbf{R}^{-1}$ or anti-phase isolation.

4.3 Computational Efficiency & Spectral Compression

- Mitigating $O(N^3)$ voxel grid costs using Spherical Harmonics (Y_l^m) coefficient arrays $O(L^2)$.

5 Real-World Physical & Biological Models

5.1 Forest Ecosystem Dynamics

- Modeling flora, fauna, soil, and moisture resource completion via vector addition ($A + B = AB$).

5.2 Targeted Viral/Pathogen Cancellation (Selective Subtraction)

- Isolating disease signatures and applying 180° anti-phase mirror vectors to achieve zero-state equilibrium ($A - B = 0$).

5.3 Atmospheric Storm Systems (Phase Shifts and Shear Turbulence)

- In-phase wave coherence vs. minor phase shift (0.01) driving localized vortices and trajectory shifts.

6 Discussion & Related Work

6.1 Comparison to Quantum Bloch Spheres and Hilbert Spaces

6.2 Integration with machine learning latent embeddings (Cosine Similarity)

6.3 Hardware Realization: Optical Phase Computing vs. Digital GPUs

7 Conclusion & Future Directions